least stable landforms, and therefore changes to their ecology happen relatively quickly. As Cowles walked among the dunes, he noticed that when certain plants died off, their decomposing matter created favorable conditions for other plants. As these new plants died, even more plants could grow.

Based on his observations, Cowles developed the idea of ecological succession, although groundwork for the concept had been laid by earlier naturalists. In an 1860 address to members of the Middlesex Agricultural Society, Massachusetts, Henry David Thoreau had stated: “Though I do not believe that a plant will spring up where no seed has been, I have great faith in a seed.”



15,000 years ago, there would only have been bare sand around Lake Michigan’s shore. Vegetation developed in a physical gradient, with sand nearest the water and forests farthest back.